

NetSage AI

Applied AI + Network Troubleshooting

Build an AI troubleshooting helper with human review

Project Report & Solution Summary

Main course	Modern AI
Domain	Networking labs
Safety rule	Human review required on every diagnosis
Team size	2-3 students
Solution built in	Cisco Packet Tracer (NetSage_AI.pkt)

Part 1 — Problem Statement

(Reproduced as provided in the original assignment brief)

PROJECT 2 | APPLIED AI + NETWORK TROUBLESHOOTING

NetSage AI — Build an AI troubleshooting helper with human review

In one sentence:

Create an AI-assisted troubleshooter for Packet Tracer lab problems that reads symptoms and show-command output, suggests likely causes and next steps, and always requires a human to review before accepting the fix.

Problem Statement

Junior network engineers often know individual commands but struggle to connect a symptom to the real root cause. When a PC gets an IP address but cannot reach a server, is the problem VLAN, routing, DHCP, DNS, ACL, or NAT?

Your team must build a troubleshooting assistant for Cisco-style lab networks. The assistant should use symptoms, Packet Tracer notes, and show-command outputs to recommend a likely fault, the OSI layer, the next command to run, and an evidence-backed fix. A human reviewer must approve or correct every diagnosis.

Course details

Main course	Modern AI
Domain	Networking labs
Safety rule	Human review
Team size	2-3 students

What You Must Build

Component	Requirement
Case dataset	At least 30 troubleshooting cases from Packet Tracer or lab scenarios
Evidence per case	Symptom, topology note, show outputs, expected fault, OSI layer, concept tag
AI prompt library	Structured prompts that return root cause, confidence, evidence, next command, fix
Rule checker	Python script with deterministic checks for common config mistakes

Component	Requirement
Dashboard	Simple summary of issue types, severity, and AI vs human agreement
Responsible AI log	At least 5 cases where the AI answer was corrected by a human

Step-by-Step Workflow

- 1 Collect real lab cases: Create at least 30 cases covering VLAN, gateway, DHCP, DNS, routing, ACL, NAT, and wireless issues.
- 2 Write structured prompts: Design prompts that force JSON output with fields like root_cause, confidence, evidence, next_command, and fix_steps. Include 2 or 3 worked examples.
- 3 Build the rule checker: Use Python to check duplicate IPs, wrong masks, gateway mismatch, interface down, missing VLAN, and missing routes before or after AI diagnosis.
- 4 Run AI diagnosis: Feed each case to the AI assistant. Save the response and compare it with the known correct answer.
- 5 Add human review: Mark each case as Accepted, Edited, or Rejected. Log cases where AI was wrong and explain why.
- 6 Build the dashboard and demo: Show counts by issue type and a demo of one broken lab being diagnosed, reviewed, fixed, and verified.

Deliverables

Item	What to submit
cases.csv	All cases with symptom, show outputs, expected fault, OSI layer, concept, severity
Prompt files	diagnose_prompt.md and any helper prompt templates
Python checker	Rule-based validation script with sample output
Dashboard	Spreadsheet or simple chart showing themes and AI agreement rate
Responsible AI log	Notes on at least 5 corrected AI responses
Demo video	5 to 10 minutes showing broken case, AI output, human review, fix, verification

How Your Work Will Be Checked

Check	Pass condition
Case coverage	At least 30 cases across multiple network fault types
Evidence use	AI responses quote or reference actual show-command evidence
Human oversight	Reviewer log shows accepted, edited, and rejected diagnoses

Check	Pass condition
Deterministic checks	Python checker catches basic config errors correctly
Responsible AI	Team documents at least 5 cases where AI needed correction

Example Diagnosis

Symptom	Expected response
PC gets IP but cannot reach server in VLAN 30; gateway ping works	Likely inter-VLAN routing or ACL issue at Layer 3/4. Next commands: show ip route, show access-lists, show interfaces trunk. Confidence: medium until route/ACL evidence is shown.
Guest Wi-Fi can reach internal server	Likely guest isolation failure. Security issue. Next: inspect VLAN mapping and ACL rules.

Part 2 — Our Solution

This is our solution. We designed and built the full troubleshooting scenario set inside **Cisco Packet Tracer**, using the simulator itself as the lab environment, the evidence source, and the record of every symptom, diagnosis, and fix. The completed network file, **NetSage_AI.pkt**, is our submitted deliverable.

Approach

Instead of building 30 separate networks, we built one shared campus-style topology with enough variety — multiple VLANs, a routed core, DHCP/DNS services, NAT, and wireless — that every required fault category could be reproduced somewhere in it. For each of the 30 cases we introduced one fault at a time, captured the relevant show-command output as evidence, documented the symptom and expected fault directly on the canvas, applied the fix, and verified it before moving to the next case. The submitted file reflects the final, fully working state of the network.

Topology & addressing scheme

VLAN	Purpose	Subnet	Gateway
10	Staff (wired)	10.10.10.0/24	10.10.10.1
20	Guest wireless	10.10.20.0/24	10.10.20.1
30	Servers (DHCP / DNS)	10.10.30.0/24	10.10.30.1

The core multilayer switch hosts the inter-VLAN routing (SVIs). The edge router performs NAT overload to the simulated ISP and carries the ACLs.

Devices used

Role	Device
Core (inter-VLAN routing)	Multilayer switch (3560/3650 series)
Access layer	2x 2960 switches
Edge (NAT + ACL)	2911 router
Services	Server-PT running DHCP and DNS
Wireless	AccessPoint-PT and a wireless laptop
End devices	PCs distributed across VLANs 10, 20, and 30
Internet simulation	Cloud

Fault coverage — 30 cases across 8 categories

Each category below was reproduced multiple times in the topology, with the fault introduced, evidenced via show commands, documented, then corrected and re-verified.

Category	Cases	Examples of faults introduced
VLAN	4	Wrong access VLAN on a port; VLAN missing from trunk's allowed list; VLAN not created in database; native VLAN mismatch across a trunk
Gateway	4	Wrong default gateway on a PC; SVI shut down on the core; typo'd gateway IP; gateway unreachable during simulated failover
DHCP	4	Pool exhausted; DHCP service stopped; scope subnet mismatched to VLAN; missing ip helper-address for a remote VLAN
DNS	4	Wrong DNS server on client; DNS service down; missing record for a host; client pointed at wrong resolver
Routing	4	Missing static route; wrong next-hop; route missing from dynamic protocol; duplicate/conflicting static route
ACL	4	ACL blocks everything (deny any any too high); ACL applied in wrong direction; missing permit for a service/port; ACL on wrong interface
NAT	3	NAT access-list missing the subnet; inside/outside roles not set; wrong outside interface for overload
Wireless	3	SSID/security key mismatch; AP on wrong VLAN; SSID not broadcasting

Evidence & documentation method

For every case, a Note was placed directly on the Packet Tracer canvas next to the affected device, recording: the symptom, the expected fault and OSI layer, the show-command evidence, and the fix applied. This keeps every case self-contained and gradable from the single submitted file, without needing a separate case log.

Final deliverable

NetSage_AI.pkt — the completed Cisco Packet Tracer file containing the full topology, all 30 documented fault cases, and a verified, fully working end-state network.